

EXP 22 – Bird Flight Checking

Name: Hemalakshmi(192571049)

Aim

To write a Prolog program that determines whether a given bird can fly or not by defining facts for birds, facts for birds that cannot fly, and rules to infer flying ability.

Algorithm

1. Define a list of birds using bird(X).
2. Define birds that cannot fly using cannot_fly(X).
3. A bird can fly if:
 - It is a bird, and
 - It is **not** listed under cannot_fly.
4. Use a rule can_fly(X) to check flying ability.
5. Use check_all to display whether each bird can fly or cannot fly.

Source Code

```
prolog
```

```
bird(sparrow).
```

```
bird(eagle).
```

```
bird(penguin).
```

```
bird(ostrich).
```

```
bird(parrot).
```

```
cannot_fly(penguin).
```

```
cannot_fly(ostrich).
```

can_fly(X) :-

bird(X),

\+ cannot_fly(X).

check_all :-

forall(bird(X),

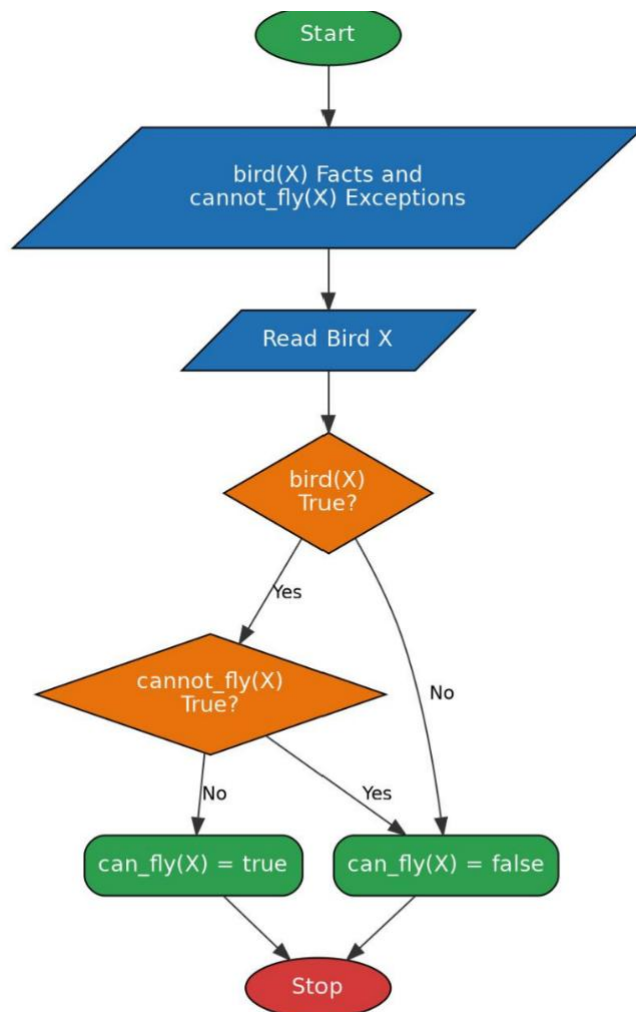
(can_fly(X)

-> format("~w can fly.~n", [X])

; format("~w cannot fly.~n", [X])

)).

Flowchart:



Output:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?- cd('C:/Users/yanar/Documents/CSA1717').
true.

?- [birds].
true.

?- check_all.
sparrow can fly.
eagle can fly.
penguin cannot fly.
ostrich cannot fly.
parrot can fly.
true.

?- can_fly(eagle).
true.

?- can_fly(penguin).
false.
```

Result

The program correctly identifies which birds can fly and which cannot by checking each bird against the cannot_fly list. It prints a clear message for every bird, showing whether it is capable of flying.